

Research on Brain-Signal-Based Communication for Non-Speaking Individuals

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SECONDARY RESEARCH -

This section examines how language is processed within the brain. Language formation involves multiple interconnected brain regions working together before speech is produced. Neural activity related to language exists even when speech is not physically expressed.

Research Paper 1

Title: The Neural Basis of Language

Link: <https://www.nature.com/articles/nrn2113>

Summary of the paper:

This paper explains how Broca's and Wernicke's areas coordinate to produce and understand language, forming the core neural system for speech processing.

Research Paper 2

Title: Brain Networks for Language Processing

Link: <https://www.frontiersin.org/articles/10.3389/fnins.2019.00133/full>

Summary of the paper:

This study shows that language is processed across distributed neural networks, where multiple brain regions interact dynamically before speech output occurs.

These studies support the idea that language can be decoded directly from neural signals.

This section focuses on understanding how the human brain processes non-verbal communication signals. Communication is not limited to speech alone; gestures and visual cues also carry meaning, which is interpreted by neural activity. Research studies have shown that specific brain responses are triggered when individuals observe communicative gestures.

Research Paper 1

Title: The Neural Basis of Non-Verbal Communication: Enhanced Processing of “Give-Me” Gestures in Infants

Link: <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2015.00059/full>

Summary of the paper:

This study used EEG recordings to analyze how the brain responds to communicative gestures. The results showed that gestures conveying intent activate specific neural patterns, demonstrating that the brain can interpret meaning without spoken language.

Research Paper 2

Title: Neural Mechanisms of Gesture Processing

Link: <https://www.frontiersin.org/articles/10.3389/fnhum.2014.00930/full>

Summary of the paper:

This research showed that observing gestures activates language-related brain regions, suggesting that non-verbal communication shares neural pathways with spoken language.

These findings establish that communication is fundamentally a neural process and can be interpreted from brain signals.

This section focuses on challenges and future directions in brain-computer interface research. Despite advancements, several technical and ethical challenges remain.

Research Paper 1

Title: Advances in Brain–Computer Interfaces

Link: <https://www.nature.com/articles/s41598-025-07438-z>

Summary of the paper:

This paper reviews recent advancements in BCI technologies and improvements in decoding accuracy.

Research Paper 2

Title: Ethical and Technical Challenges in BCI

Link: <https://www.sciencedirect.com/science/article/pii/S1364661322002465>

Summary of the paper:

This study highlights challenges such as signal variability, privacy concerns, and ethical implications.

These studies outline limitations and guide future research directions.

This section focuses on EEG signal theory and its role in measuring brain activity. EEG is widely used to capture electrical signals generated by neurons, but these signals are complex and require careful processing.

Research Paper 1

Title: EEG Signal Processing Review

Link: <https://www.mdpi.com/1424-8220/17/9/1937>

Summary of the paper:

This paper discusses EEG signal characteristics, including noise, artifacts, and preprocessing methods such as filtering and feature extraction.

Research Paper 2

Title: EEG-Based Brain–Computer Interfaces

Link: <https://www.sciencedirect.com/science/article/pii/S1388245718302913>

Summary of the paper:

This study explains how EEG signals are used in BCI systems and highlights challenges such as low resolution and variability across users.

These findings provide the theoretical basis for using EEG in brain-based communication systems.

This section focuses on decoding speech directly from neural activity. Research has shown that brain signals associated with speech planning can be translated into text or audio.

Research Paper 1

Title: A High-Performance Speech Neuroprosthesis

Link: <https://www.nature.com/articles/s41586-023-06377-x>

Summary of the paper:

This study demonstrated that neural signals can be decoded into text in real time for individuals with paralysis, enabling communication without speech.

Research Paper 2

Title: Direct Speech Synthesis from Brain Activity

Link: <https://www.nature.com/articles/s41586-019-1119-1>

Summary of the paper:

This research showed that neural activity can be converted into synthesized speech, proving that brain signals contain detailed speech information.

These studies validate the feasibility of brain-based communication systems.

This section examines the architecture of brain-computer interface systems. BCI systems follow a structured pipeline that converts neural signals into usable outputs.

Research Paper 1

Title: Brain–Computer Interface Systems

Link: <https://www.frontiersin.org/articles/10.3389/fnins.2020.00025/full>

Summary of the paper:

This paper explains the components of BCI systems, including signal acquisition, preprocessing, classification, and output generation.

Research Paper 2

Title: Design of Communication Systems Using BCI

Link: <https://ieeexplore.ieee.org/document/8716935>

Summary of the paper:

This study discusses system design considerations for BCI-based communication systems and integration challenges.

These studies define the structure required to build a brain-signal communication system.

This section focuses on machine learning techniques used to decode EEG signals. Neural signals are complex and require advanced models for accurate interpretation.

Research Paper 1

Title: EEG-Transformer: Self-Attention for Imagined Speech Decoding

Link: <https://arxiv.org/abs/2112.09239>

Summary of the paper:

This research used transformer models to decode imagined speech from EEG signals, improving classification accuracy using attention mechanisms.

Research Paper 2

Title: Deep Learning for EEG Signal Analysis

Link: <https://www.sciencedirect.com/science/article/pii/S0893608020301056>

Summary of the paper:

This paper explains how deep learning models such as CNNs can extract meaningful features from EEG data.

These studies demonstrate how AI enables interpretation of brain signals.

This section focuses on implementing brain-computer interfaces using embedded systems. Low-cost hardware platforms enable practical development of brain-based communication systems.

Research Paper 1

Title: PEEEG: Raspberry Pi EEG System

Link: <https://arxiv.org/abs/2201.02228>

Summary of the paper:

This study presents a low-cost EEG system using Raspberry Pi for real-time signal acquisition and processing.

Research Paper 2

Title: Embedded Systems in Brain–Computer Interfaces

Link: <https://ieeexplore.ieee.org/document/8978836>

Summary of the paper:

This paper discusses how embedded platforms can be used to implement portable BCI systems.

These studies demonstrate the feasibility of building practical systems.

PRIMARY RESEARCH -

Interviewer: Could you share your experience caring for someone with communication difficulties?

Caregiver: Yes, my brother has ALS, and over time he has gradually lost his ability to speak.

Interviewer: How does he currently communicate?

Caregiver: He mostly uses eye movements and sometimes a communication board, but it's quite slow.

Interviewer: Is that method effective for daily communication?

Caregiver: It works for basic needs, but it's very limited when he wants to express something more detailed.

Interviewer: Does this situation cause frustration?

Caregiver: Yes, definitely. He gets frustrated when we don't understand him quickly, and it's hard for us too.

Interviewer: Have you tried any assistive technologies?

Caregiver: Yes, we tried an eye-tracking device, but it requires a lot of focus and can be tiring.

Interviewer: What do you think about a system that converts brain signals into text?

Caregiver: That sounds really helpful, especially since he cannot move much now.

Interviewer: What features would be most important for you?

Caregiver: It should be accurate, fast, and easy to use without too much effort.

Interviewer: Do you have any concerns about such a system?

Caregiver: Yes, reliability is important. It should work consistently without errors.

Interviewer: Do you think he would be comfortable using an EEG headset?

Caregiver: If it is lightweight and not complicated, I think he would be willing to try it.

Interviewer: Final thought?

Caregiver: If this system works well, it could really improve his independence and quality of life.

Interviewer: Could you share your experience caring for someone with speech difficulties?

Caregiver: Yes, my mother had a stroke, and since then she has trouble speaking clearly.

Interviewer: How does she usually communicate?

Caregiver: She mostly uses hand gestures and sometimes tries to write things down.

Interviewer: Is that method effective?

Caregiver: Not really. It takes a lot of time to understand what she is trying to say.

Interviewer: Does this create frustration?

Caregiver: Yes, a lot. She gets upset when we don't understand her, and it's stressful for us too.

Interviewer: Have you tried any assistive devices or apps?

Caregiver: Yes, we tried one app, but it was slow and a bit difficult for her to use.

Interviewer: What do you think about a system that converts brain signals into text?

Caregiver: That would be very helpful, especially if it works quickly.

Interviewer: What features would be most important to you?

Caregiver: Accuracy and speed are the most important.

Interviewer: Do you have any concerns about such a system?

Caregiver: Yes, mainly cost and whether it will work reliably every time.

Interviewer: Would your mother be comfortable using something like an EEG headset?

Caregiver: If it is simple and comfortable, I think she would be willing to try it.

Interviewer: Final thought?

Caregiver: If it works properly, it could make daily communication much easier for us.

Interviewer: Thank you for taking the time to speak with us. Could you describe your situation?

Caregiver: Yes, I take care of my grandmother. She has difficulty speaking clearly due to age-related neurological issues.

Interviewer: How does she communicate her needs?

Caregiver: She mostly uses simple words, gestures, and sometimes points at things around her.

Interviewer: Is that method effective for daily communication?

Caregiver: It works for very basic needs, but anything more complex becomes difficult to understand.

Interviewer: Do misunderstandings happen often?

Caregiver: Yes, quite often. Sometimes we think she is asking for one thing, but she actually means something else.

Interviewer: How does that affect her emotionally?

Caregiver: She gets upset and sometimes withdraws from conversations because it becomes too tiring.

Interviewer: Have you tried any communication tools or technologies?

Caregiver: We tried a simple communication board, but she found it confusing and difficult to use regularly.

Interviewer: What do you think about a system that converts brain signals into text?

Caregiver: That would be very helpful, especially if it reduces the effort she needs to communicate.

Interviewer: What features would you expect from such a system?

Caregiver: It should be very simple, accurate, and not require too much setup.

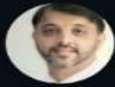
Interviewer: Do you have any concerns about this technology?

Caregiver: Yes, mainly whether it will work consistently and if it will be affordable.

Interviewer: Finally, do you think this system could improve her quality of life?

Caregiver: Yes, definitely. It would help her express herself better and feel more independent

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- 1) https://drive.google.com/file/d/1TJ8JXBdJgb8_5Ekgo9r5lCtmEjHLi0yx/view?usp=drivesdk
 - 2) https://drive.google.com/file/d/1Z-0t4FZXX0qMfrSFM77xC_A3hJk8M-mZ/view?usp=drivesdk
 - 3) https://drive.google.com/file/d/1gp8TIEcbMI-eW71Bhg17sgllfb13mT5_/view?usp=drivesdk
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Amir Uncle



Hi uncle...need a slight interview if u are free

1:07 pm ✓✓

Yeah beta, sure. Im free now

1:17 pm

Thank you soo let's start

1:17 pm ✓✓



Soo this interview is about non speaking patients

1:17 pm ✓✓

So could you please tell me your experience with the patients who can't speak

1:18 pm ✓✓

Yeah, Ive worked with patients who have ALS and strokes. And many of them cannot speak even though they have clear thoughts

1:20 pm

So their thinking ability remains intact?

1:20 pm ✓✓

Yes

1:20 pm

In most cases their cognition is preserved

1:21 pm

But motor control for speech is impaired

1:21 pm

So how do they communicate currently

1:21 pm ✓✓

Mostly through

1:21 pm

Eye tracking systems

1:22 pm



Message





Amir Uncle



Mostly through

1:21 pm

Eye tracking systems

1:22 pm

Gestures or communication boards

1:22 pm

So do you think these methods are effective

1:22 pm ✓✓

I mean, they work

1:22 pm

But they are slow and require a lot of effort

1:22 pm

What emotional impact does this have

1:24 pm ✓✓

A lot of frustration

1:24 pm

And sometimes even depression

1:24 pm

So we are proposing a brain signal based communication system

1:25 pm ✓✓

What do you think about this

1:25 pm ✓✓

Oh wow

1:25 pm

That sounds promising

1:25 pm

There is a research on imagined speech I believe

1:26 pm

That supports your idea

1:26 pm

Oh yaa i know it

1:26 pm ✓✓



Message





Amir Uncle



So what challenges do you foresee 1:26 pm ✓✓

Oh there are a lot of things 1:27 pm

But the main things that come to mind are
1:27 pm

Signal noise 1:28 pm

Variability between patients 1:28 pm

Yea these I can think of so far 1:28 pm

So would patients accept such technology
1:28 pm ✓✓

It is a very innovative idea and I do think it
would be comfortable and reliable if done
correctly 1:29 pm

So what about your final thoughts 1:29 pm ✓✓

This would be revolutionary and very hard to
achieve but a great research topic 1:30 pm

It would help a lot of people Edited 1:30 pm

Ok thanks...and I need your consent to share
these chat messages as part of my research
1:31 pm ✓✓

Yea sure go ahead 1:31 pm

Ok thank you 1:31 pm ✓✓



Message





Yashoda Ent Doctor



Hi sir can I speak 2 min 1:37 PM ✓✓

Hello 1:52 PM

You

Hi sir can I speak 2 min

Yes sure 1:52 PM

Can I ask some questions for my interview 1:53 PM ✓✓

And my topic is about problem faced by non speaking patient 1:53 PM ✓✓

Okay sure 1:54 PM

Can you describe how patients with speech difficulties communicate? 1:54 PM ✓✓

They mostly rely on gestures 1:54 PM

Boards 1:54 PM

Or some assistive devices 1:55 PM

Usually what are the limitations? 1:55 PM ✓✓

They require a lot of effort 1:56 PM

And are mainly slow 1:56 PM

Do patients get tired? 1:56 PM ✓✓



Message





Yashoda Ent Doctor



Do patients get tired? 1:56 PM ✓✓

Yes 1:56 PM

Pretty quickly 1:57 PM

What do you think about brain-based systems, do you think they will help?

1:58 PM ✓✓

Yeah they would definitely help for severe cases, but it would be tough to achieve

1:59 PM

You mentioned it would be tough to achieve, what are the challenges we would face?

2:00 PM ✓✓

Yeah it would be mainly in the training and adaptations

2:00 PM

It would require a lot of work as well, and advanced technology

Edited 2:01 PM

What features are important to implement?

2:02 PM ✓✓

Ease of use mainly 2:02 PM

And accuracy 2:03 PM

Do patients need personalization?

2:03 PM ✓✓

Yes definitely 2:03 PM



Message



←  Yashoda Ent Doctor   

adaptations

2:00 PM

It would require a lot of work as well, and advanced technology

Edited 2:01 PM

What features are important to implement?

2:02 PM ✓✓

Ease of use mainly

2:02 PM

And accuracy

2:03 PM

Do patients need personalization?

2:03 PM ✓✓

Yes definitely

2:03 PM

Each patient is different

2:03 PM

Thank you for the interview any final opinion??

2:04 PM ✓✓

It has great potential but equally high effort and work

2:04 PM

You'll need to do rigorous training

2:04 PM

Ok thanks and I need your consent to share these chat messages as part of my research

2:06 PM ✓✓

Yea sure

2:06 PM

Ok thank you

2:06 PM ✓✓



 Message



How are you currently communicating with others?

Is writing a comfortable method for you?

Are your family members able to understand you easily?

Does that cause frustration for you?

Do you feel that slow communication affects your daily life?

If there was a system that converts your brain signals into text, would you be willing to use it?

Would you be comfortable wearing an EEG headset?

What matters more to you speed or accuracy?

Would you be okay with spending sometime learning or training to use the system?

Finally, how do you think such a system could impact your life?

I mostly use gestures, and sometimes write things down. But it ain't very easy for me.

Not really. My hands feel stiff. It's too much effort

Sometimes they do but half the times, then misunderstand.

Yes, saying something ^{simple} takes a lot of time and effort. Makes me feel frustrated

Absolutely, simple things are hard to get through, makes life harder.

Yes, that would be so nice to have something like that man.

Yeah, no problem, given that it's lightweight and comfortable.

Accuracy, any day.

Yes, if it promises results

It would be a life changer, I mean it's gonna be real nice, having not to struggle getting out simple thoughts, saves a lot of time and effort.

1/1
आप अभी कैसे संवाद करते हैं?

मुख्य इशारे और थोड़ा लिखने से करता हूँ।

क्या यह आसान है?

नहीं काफी श्लो है

आपका निराशा होती है?

हां अब लीम समझ नहीं पाते

अगर बिना बोलें कम्युनिकेशन कैसे रहेगा?

बहुत अच्छा होगा

गति या सटीकता - क्या महत्वपूर्ण है?

दोनों, लेकिन सटीकता ज्यादा।

प्रशिक्षण करने में समस्या?

थोड़ा साइम लगेगा तो भी ~~भी~~ चलेगा

क्या ये हैमि उपयोग करेंगे?

हां जरूर

अंतिम अनुभूति?

ये सिस्टम मेरी लाइफ बदल सकता है

Interviewer: Thanks for talking with me. Can you tell me how you usually communicate with others?

Patient: I mostly use gestures and facial expressions. Sometimes I try a communication board, but it's not always easy to use.

Interviewer: Do those methods let you express everything you want to say?

Patient: Not really. I can only communicate basic things. It's very hard to share detailed thoughts or feelings.

Interviewer: How does that make you feel emotionally?

Patient: It's really frustrating. I often feel stuck because I know what I ~~need~~ want to say, but I just can't express it properly.

Interviewer: Have you tried any assistive communication apps or devices?

Patient: Yes, I tried a mobile app once, but it took too much effort and was slow to use.

Interviewer: What are the biggest problems with current methods, mainly the time and effort. Everything feels slow and it gets tiring after a while.

Interviewer: We're working on a system that converts brain signals into text. What do you think about it?

Patient: That sounds really helpful. If it works well, it could make communication much easier for me.

Interviewer:- would you be open to using something like an EEG headset regularly?

That sounds really helpful. if it works well, it could make communication much easier for me. yes, as long as it's comfortable.

Interviewer:- what features would you want in such a system?

yes, as long as it's comfortable and not too complicated to set up. It should be fast and accurate.

Interviewer:- Do you think learning to use it would be difficult?

It should be accurate, fast, and easy to use. I don't want something that makes things more complicated.

Interviewer:- Finally, how do you think this system could change your daily life?

It would give me more independence i wouldn't have to rely on others to understand what i need all the time.